

What is worth inheriting: a measurement platform for transferred operating experience in technical systems

Evgenii Bogomazov-Pusser · independent researcher · evgeniipusser88@gmail.com

Draft, September 2026. All figures reproduce from open code and public data.

Abstract

A newly commissioned building, machine or network has no operating history of its own. The question here is not whether experience can be transferred between such objects. It can, and methods exist. The question is what is worth transferring, and how that can be measured instead of asserted.

We describe a measurement platform. An individual object lives once, without replay. Losses accumulate as they occur. The object starts either from a generic default, from the record of one predecessor, or from a fund aggregated over many predecessors. The protocol was applied in three domains — a synthetic population of buildings, measured electricity records of 40 real non-residential buildings, and six benchmark water distribution networks — and to five candidate inherited objects.

The paper is written as the chain of experiments in the order they were performed. It includes three defects of the instrument found after the experiments were finished, what each defect changed, and one reported figure that a controlled repetition did not support and which is withdrawn. Five results hold across domains. First, a complete inherited representation applied unchanged is usually worse than what the receiver can obtain for itself, even from two weeks of its own history. Second, similarity of bodies does not predict whether transfer helps; an apparent correlation of +0.21 in the first experiment turned out to be an artefact of the cost function and fell to +0.05 once the artefact was removed. Third, what transfers is the arrangement of observation, not the values: the rule for placing sensors transfers where the detection rule does not. Fourth, the quality a predecessor achieved does not pass to its successors; across two populations the slope of successor gain on predecessor quality was -2.12 and -0.38 , negative in sign and undetermined in size. Fifth, most of the harm attributed to inheritance is an artefact of the units in which records are written; the same records expressed in quantities defined relative to the receiver raise the share of objects helped from 81–84 % to 99 %.

A separate measurement bounds all of these. For each object we compute the best any schedule method could achieve, and find that four objects in five have almost nothing to gain while one in five has 11–41 %. What an object gains from the fund is closely related to how much it had to gain, with a correlation of +0.91, so the fund can be read as a ranking of which objects are worth attention — a question no object can answer about itself. We end with what the platform does not measure, including a measurement of its own range which shows that range to be narrow.

1. Introduction

Technical objects that adapt during their operating life are now common. Buildings whose controls follow occupancy. Networks whose monitoring adjusts to observed failures. Fleets whose maintenance is revised as evidence accumulates. Each object acquires, over years, information specific to it and expensive to obtain. When it is decommissioned or rebuilt, that information is lost, and its successor starts from generic settings.

We begin with buildings, because buildings are the best documented of these objects. Public hourly records exist for thousands of them. Benchmark networks ship with the simulation software. The same questions arise for machines, factories and vehicles, and the platform is not specific to buildings, but the material is here.

Transfer between buildings is an active field and not a new idea. Xu et al. [1] transfer the strategy part of a reinforcement-learning controller and retrain a building-specific physical part locally, obtaining stable control from the first day. A critical review of transfer learning in buildings [2] treats negative transfer as a recognised phenomenon, with its causes and mitigations an open subject, and identifies the selection of source buildings as an unsolved part of the problem. A dataset of 15 808 building models was released in 2025 specifically for transfer research in building control [3].

The open question is prior to method. Given an object with an operating history, what part of that history is worth preserving so another object can use it? The answer is usually assumed rather than measured. In evolutionary robotics the inherited object is fixed in advance — a genotype, with lifetime learning written back into it [14]. In building control it is a set of controller parameters or a forecasting model. In each case the question of what to inherit has been settled before the experiment begins. We treat it as an empirical question and supply an apparatus.

Two commitments shape the apparatus. An individual lives once. A building experiences each winter one time, cannot re-run it, and pays for mistakes in comfort and energy. Any protocol that lets the receiver replay its own life makes self-acquisition artificially cheap and inheritance artificially worthless. And the measured quantity is loss accumulated during the first life, not the quality of the final model, because the value of inherited experience is exactly the cost of acquiring it again.

The paper follows the experiments in order. Two of them rested on a wrong premise. Three defects of the instrument were found later. All are reported with the result that exposed them. This is not a stylistic choice. Each time the instrument reported that inheritance was worthless, the cause was the design rather than the phenomenon, and that is the strongest evidence available that the instrument measures something.

2. The platform

2.1 Elements

A measurement requires four elements from any candidate domain.

Element	Meaning	Buildings	Water networks
Body	fixed properties of the	ten thermal and use	the pipe graph,

Element	Meaning	Buildings	Water networks
	individual	parameters	junctions, elevations
Life	a single pass through time, no replay	hourly simulation of one or more years	45 scenarios of three days each
Loss	what is paid while living	energy cost plus discomfort	missed leaks and false alarms
Record	what an individual leaves behind	a setpoint schedule and its conditions	a placement rule or a detection rule

Table 1. The four elements required of a domain.

2.2 Conditions compared

Every experiment compares the same starting conditions. Every figure is a percentage change in accumulated loss relative to the first of them.

- **alone** — the receiver starts from a generic default and learns only from its own single life;
- **one predecessor** — the receiver starts from the record of a single earlier individual;
- **fund** — the receiver starts from an aggregate over many earlier records;
- **fund, refined** — the same, followed by the receiver's own short adaptation.

Positive numbers mean the inherited start was better than starting alone. The protocol declares before each run what counts as confirmation and what as refutation. Negative outcomes are reported.

3. Step 1: transfer with replay, and a cost function that moved

Ten synthetic buildings differ in ten parameters: envelope loss, thermal capacity, effective glazing, heating power, infiltration, internal gains, occupancy hour, mean outdoor temperature, daily swing, orientation. Each simulates a season of 60 days at hourly resolution. Cost combines energy and degree-hours of discomfort during occupancy. A controller is either five parameters — occupied setpoint, unoccupied setpoint, preheating lead, deadband, gain — or a schedule of 24 hourly setpoints. A donor has 400 evaluations of a season, a newcomer 40. Three populations give 270 ordered pairs per experiment.

Condition	Five parameters	Twenty-four setpoints
inherited as is	−11.9 % (helped in 26 % of pairs)	−3.6 % (helped in 52 %)
inherited as starting point	+3.8 % (helped in 81 %)	+0.5 % (helped in 63 %)
correlation of gain with body distance	+0.05	≈ 0

Table 2. Step 1, after the cost function was repaired. Transfer between simulated buildings under replay-based learning. 270 ordered pairs per column.

Choosing the donor did not help. A linear rule predicting gain from per-parameter differences, fitted on two populations and tested on a third, gave +4.7 % against +4.1 % for a random donor. Choosing the nearest body by unweighted distance gave +3.5 %, worse than random. The best donor in hindsight gave +5.4 %.

The numbers in the table are not the ones first obtained. The original cost function counted discomfort below the occupied setpoint, and the occupied setpoint was one of the five numbers the building was

tuning. A building could lower its promise to the occupants from 21 °C to 18 °C and thereby move the threshold for the penalty from 20.5 °C to 16 °C. In one run this is exactly what it did, cutting cost from 3 078 to 1 915 mostly by deciding that 16 °C was acceptable. The building was not learning to heat more cheaply. It was moving the ruler by which it was measured.

The comfort threshold was fixed at 21 °C and Step 1 was re-run on the same three populations. The main result survived: an inherited setting used whole is harmful, and the same setting used as a starting point helps. One thing changed. The correlation of gain with body distance fell from +0.21 to +0.05, that is to nothing. The only number in the original Step 1 that suggested body similarity might matter was produced by the moving threshold.

There is a second, larger problem with Step 1, and it is not fixable by patching. A newcomer allowed 40 replays of the same season reaches most of what a donor reached in 400, because in this setting a season can be re-lived at no cost. No building can do this. The design was discarded.

4. Step 2: one pass, losses accumulated

The protocol was rebuilt around a single life. Each building lives year after year in hourly simulation. Once a week it may try one modified schedule. The week is lived once, its cost is paid, and the modification is kept only if the week compared favourably with a regression of past weeks on mean outdoor temperature. Predecessors, having lived and kept a log, tune a schedule offline on their own record — the counterpart of an old building with years of data and a calibrated model. Newcomers begin from a generic 21 °C schedule, from a predecessor's tuned schedule, or from the median of ten such schedules.

The experiment was run twice with one difference. In the first version the year is ordinary. In the second a rare condition may occur: a severe winter, a heat episode, or a heating plant at half capacity. Eight predecessors, ten newcomers, the same buildings and the same weather in both versions, the same tuning budget of sixty evaluations.

Because the integration step is a known weakness of this domain (§10), the whole experiment was run twice more, at one hour and at ten minutes, with everything else held identical. All four cells are reported.

Year	Step	One predecessor	Median over predecessors
ordinary	one hour	−0.4 % (helped 50 %)	+0.6 % (helped 80 %)
ordinary	ten minutes	−2.8 % (helped 20 %)	+0.1 % (helped 60 %)
rare conditions	one hour	−4.5 % (helped 0 %)	+0.2 % (helped 90 %)
rare conditions	ten minutes	−3.9 % (helped 0 %)	+0.4 % (helped 100 %)

Table 3. One pass through time. Ten newcomers per cell, identical buildings and weather across cells; only the year type and the integration step differ.

Two things hold in every cell. An inherited schedule taken from a single predecessor is harmful. A median over many predecessors is not.

Rare conditions sharpen the first of these and leave the second. A single predecessor goes from helping half the newcomers in an ordinary year to helping none at all when rare conditions are present, at both step sizes. It survived its own rare event, not the receiver's. The fund, which contains all the cases, helps 90 % and 100 % of newcomers in the two rare cells.

The magnitudes are small: between +0.1 % and +0.6 % for the fund. An earlier version of this experiment, with a different tuning budget and a different set of buildings, reported +5.5 % for the fund in the rare condition. That figure did not survive the controlled repetition and is withdrawn. What survives is the direction and the share of objects helped, not the size of the gain.

Ten newcomers per cell. The ordering is consistent across four cells; the magnitudes are not established.

5. Step 3: measured buildings

The Building Data Genome Project 2 [4] provides hourly electricity for 1 636 non-residential buildings over 2016–2017, with weather and metadata. We use site "Rat": 40 buildings with at least 95 % coverage. Each building's load shape — consumption divided by its own mean — is predicted from hour, day of week, month, weekend, outdoor temperature and its lag. Models are gradient-boosted. 2016 trains, 2017 tests. A newcomer is a building for which only the first 14 days of 2016 exist. 1 560 ordered pairs.

Condition	Median change	Helped in
a donor's full model, receiver has full history	−85 %	7 % of pairs
a donor's model as is, receiver has 14 days	−33 %	17 % of pairs
a donor's model refined on 14 days	−21 %	28 % of pairs
best donor in hindsight	+2.5 %	52 % of buildings
model trained on all other buildings	−10.7 %	40 % of buildings
that model refined on 14 days	+2.5 %	62 % of buildings

Table 4. Measured buildings. Change relative to the receiver's own model, 1 560 ordered pairs.

The ordering of Step 2 reappears on measured data, and it is sharper. A whole inherited representation loses 85 % of accuracy, against roughly 10 % in simulation. Population knowledge combined with the receiver's own short history is again the only condition that helps a majority.

Metadata similarity is again uninformative. Correlation of gain with difference in log floor area is −0.001. Correlation with difference in year of construction is −0.06. Buildings of the same declared use lose 82 % where buildings of a different use lose 86 %. A hospital and a school of the same size are no closer to each other than a hospital and a warehouse.

62 % is not "it works". It means the inherited start helps more often than it harms. For 38 % of buildings it remained harmful, and we found no property that separates them in advance.

6. Step 4: a domain whose body is a graph

Six benchmark water distribution networks — Net1, Net2, Net3, ky4, Anytown, CCWI17-Herman-Mahmoud, from 9 to 959 junctions — are simulated with EPANET [5] through WNTR [6]. Each receives 45 three-day scenarios, 25 containing a leak of random size at a random junction. Three pressure sensors observe the network. Performance is average precision of leak detection on held-out scenarios.

6.1 Transferring the detection rule

Detection features are permutation-invariant summaries of standardised pressure deviations, so a rule fitted on one network applies to another regardless of size.

Network	Own rule, 1 failure	One donor's rule	Fund, refined
Net1	0.990	0.958	1.000
Net2	0.986	0.902	0.993
Net3	0.863	0.837	0.880
ky4	0.959	0.886	0.959
Anytown	0.987	0.312	0.983
Herman	0.205	0.211	0.206

Table 5. Average precision of leak detection. Transferring the detection rule.

A single failure suffices. The newcomer reaches 0.99 of 1.0 from one leak, and there is nothing left to inherit. Anytown shows the risk instead: a donor's rule gives 0.312 where the network's own rule gives 0.987.

6.2 Transferring the rule of placement

The inherited object is moved one level up: not how to decide that a leak is present, but where to watch. A placement rule assigns weights to structural properties of a junction — degree, hops from the nearest source, elevation, base demand, mean adjacent diameter — standardised within each network, and selects the three highest-scoring junctions. No node identity crosses between networks.

Network	Random	Own rule, 2 failures	One donor	Fund
Anytown	0.117	0.212	0.173	0.215
Net3	0.931	0.967	0.939	1.000
ky4	0.843	0.841	0.842	0.858
Net1, Net2	placement barely matters; all methods coincide			
Herman	0.290	0.290	0.290	0.290

Table 6. Average precision of leak detection. Transferring the rule of placement.

In the networks where placement matters, the inherited rule equals or exceeds what the individual can acquire for itself. This is the first case in the series where inheritance was necessary rather than convenient. The difference from §6.1 is the number of failures required. A detection rule needs one. A

placement rule needs dozens, because to judge whether a position is good one must see many different leaks. A new network does not have dozens and will not have them for years.

Herman stays at 0.290 under every placement. Three pressure sensors cannot see a leak in that network. This is a limit of the formulation, not of transfer.

A correction is recorded here. We first took the transferability of a placement rule to be new. It is not. Placement by topological rules and by centrality measures is established in water distribution research [7,8]. The contribution is narrower: not a new way to place sensors, but a measurement of what an inherited placement rule is worth to an object with no failures of its own.

7. Step 5: generations, conditions and rare events

Twenty objects per generation, ten generations, each object living one year and contributing one record. Rare conditions occur with fixed probability: severe winter 10 %, heat episode 8 %, heating plant at half capacity 7 %. Three funds are compared: the median of all records; a ridge regression from an object's conditions to a schedule; and the median of records from the same rare condition. The fund is frozen within a generation.

Records in the fund	Median of records	Conditions → schedule	Same condition only
up to 20	+0.8 %	+0.5 %	+1.7 %
20–60	+1.1 %	+2.3 %	+1.3 %
60–100	+1.6 %	+3.3 %	+1.9 %
100–140	+1.5 %	+4.4 %	+1.9 %
140–200	+2.2 %	+4.6 %	+2.5 %

Table 7. Value of the fund as it accumulates. Ten generations, twenty objects each.

Quantity alone helps a little. The plain median improves from +0.8 % to +2.2 %. The conditional rule is worse than the median at twenty records, because the relations are not yet visible, then overtakes it and reaches +4.6 %. Relations between conditions and good settings must be paid for in records before they can be used.

The growth is not unbounded. Between 20 and 100 records the conditional rule adds almost three percentage points. Between 100 and 200 it adds 0.2. The curve flattens, and it flattens in the same place on two further populations. The first hundred records are worth more than the next hundred. For anyone planning to collect this material, that is the practical point: there is no need to wait for thousands of buildings, and no reason to expect a thousand to give ten times what a hundred gives.

Condition of the receiving object	Median fund	Conditions → schedule	Same condition only	n
ordinary year	+1.4 %	+3.3 %	+1.9 %	147
severe winter	+0.8 %	+4.1 %	+1.1 %	20
heat episode	+1.6 %	+4.3 %	+1.8 %	25
heating at half capacity	–2.7 %	–14.8 %	–0.4 %	8

Table 8. Rare conditions. Gains relative to a generic start.

The last row matters. Climate extremes are handled, because climate is part of the recorded conditions. A damaged object is not. Heating capacity is recorded as a nameplate value while the object runs at half power, so the conditional rule extrapolates confidently and wrongly and loses 15 %. Accumulation does not reduce this harm. It increases it, because a larger fund makes the rule more confident without making it correct. A record must carry the state of the object, not only its specification.

Eight cases. The sign is worth acting on. The number is not a result and should not be quoted as one.

8. Step 6: the language in which records are written

All transfers so far moved a schedule expressed in degrees. A schedule in degrees is meaningful only relative to a particular body. The same 18 °C night setpoint is a mild setback in one building and an unreachable target in another.

We re-expressed the identical records in four dimensionless quantities defined relative to the receiving object: the depth of setback below its own comfort temperature, the lead time before its own occupancy hour, any daytime reduction, and the duration of the reduced period. A schedule for any object is reconstructed from these four numbers.

Records in the fund	Degrees	Dimensionless
20	+1.6 %	+2.9 %
40	+2.8 %	+4.1 %
80	+3.6 %	+5.5 %
120	+4.1 %	+6.1 %
all objects, population 1	+3.3 % (helped 84 %)	+4.6 % (helped 99 %)
all objects, population 2	+2.8 % (helped 83 %)	+4.8 % (helped 99 %)

Table 9. The same records transferred in two representations.

The median gain rises by about half. The decisive figure is the share of objects helped: 99 % against 83–84 %. Most of the harm previously attributed to inheritance was an effect of representation. A record written in quantities that refer to the receiver almost never damages it.

The same experiment measures the distance from each newcomer to the nearest record in the fund. Gain falls as the distance grows, from +4.1 % to +1.8 % in degrees, and falls more gently in the dimensionless form, from +5.6 % to +3.6 %. A common language is needed exactly where no similar case exists.

Records written by different objects are like two combs with different teeth missing. Nothing lines up until both are described in the same terms.

9. Step 7: is quality heritable?

Every experiment so far asked whether a fund helps. This one asks a different question. If a predecessor tuned itself well, do its successors do better than the successors of a mediocre predecessor?

Twelve predecessors tune themselves offline. Each contributes five successors. We measure the slope of successor gain on predecessor quality. A slope of 1.0 would mean that every percentage point won by the predecessor passes to the successor. Two independent populations.

	Population 1	Population 2
predecessor quality, mean	5.1 %	5.6 %
predecessor quality, spread	3.5	7.7
successor gain from one predecessor	+0.4 % (helped 55 %)	−0.2 % (helped 43 %)
successor gain from the fund	+3.6 % (helped 97 %)	+2.1 % (helped 98 %)
slope	−2.12	−0.38
correlation	−0.27	−0.16

Table 10. Heritability of quality. Twelve predecessors, five successors each, two populations.

The slope is negative in both populations. Its size differs by a factor of five, and at sixty points neither correlation is distinguishable from zero. The statement that survives is that predecessor quality does not predict successor gain. The statement that does not survive is that a good predecessor is harmful, which is what −2.12 alone appeared to say.

This explains why a fund works and a single predecessor does not. If quality were heritable, finding the best predecessor would be enough. One predecessor helps roughly half of successors regardless of how good it is. A fund helps 97–98 %, and not because it contains good predecessors but because it contains many different cases.

For the general question, this is the working answer of the series: what is inherited is not what a parent acquired, but what many have survived.

10. What the instrument got wrong about itself

Three defects were found after the experiments were complete. Two were repaired. One is declared and stands.

The building does not learn. In the one-pass protocol a building tries a modified schedule about forty times a year and accepts two or three. Measured against an untouched 21 °C schedule across thirty buildings in three populations, a year of this gives a median of −0.2 %, and helps 13 % of buildings. The mechanism does not work.

This was not repaired, for two reasons. It is closer to a real building than a working mechanism would be: schedules in real buildings are set once at commissioning and stay, and savings from a re-commissioning visit decay to about 65 % after four years as operators adjust settings in response to complaints [9]. And it changes the reading of the earlier tables rather than invalidating them. The condition called "alone" was never a self-teaching building. It was an untouched schedule. That happens to be the right comparison, but the sentence "each building learns its own easily, therefore inheritance is not needed" must be withdrawn. The building learned nothing.

The time step is too coarse. The thermal integration advances one hour at a time with the heater at full or zero output. A building needing three degrees receives 6.31 in one hour, then shuts off and falls back:

18.00 → 24.31 → 18.22 → 24.48 → 18.35 → 24.58 → 18.43

The building never holds 21 °C. At a ten-minute step the same building settles at 20.52 °C and stays there, which confirms the oscillation as an artefact of the step size rather than a property of buildings.

We first declared this defect without measuring what it costs, which is not good enough. The experiment of §4 was therefore repeated at both step sizes with everything else held identical, and the four cells are in Table 3. The ordering of conditions is the same at both steps, in both the ordinary and the rare year. What the coarse step does change is magnitude: absolute costs are inflated, and one figure reported in an earlier version of §4 did not survive and has been withdrawn.

The defect therefore stands, with its effect on the comparisons now measured rather than assumed. The remaining experiments have not been repeated at ten minutes; on the evidence of §4 their orderings should hold and their magnitudes should not be relied on.

The cost function moved. Described in §3. Present only in Step 1, repaired, Step 1 re-run.

The three are of different kinds. The third was an error in the measuring device and produced a false positive, the correlation of +0.21. The second is a known approximation carried too far. The first was a wrong belief about the device, held by the author, and stated twice in earlier drafts.

11. How much is there to take at all

A separate measurement bounds everything above. For each building we searched for the cheapest schedule using 400 evaluations of its own year with full knowledge of that year in advance, and compared it with an untouched 21 °C schedule. No method that changes only the schedule can beat this number: not self-learning, not a fund, not an engineer.

Measure	Across 30 buildings
median	+2.5 %
mean	+6.4 %
buildings above 5 %	40 %
buildings above 10 %	20 %
buildings above 15 %	13 %

Table 11. The upper bound on what any schedule method can deliver, measured per building.

The distribution matters more than the median. For four buildings in five there is almost nothing to take, and no method can help them. For one in five there is 11 % to 41 %, and the building took none of it. In the extreme case the available gain was 41 % and the building's own year of trials returned -3.2 %.

The schedule found for that building was checked on three other years of the same building and returned +44.5 %, +44.3 % and +45.7 % against +45.9 % on the year it was fitted. It is a property of the building, not of one year's weather.

Two things follow. Arguing about inheritance without this number is arguing about nothing, because for most buildings every condition is equivalent by construction. And the useful product of a fund may not be a better schedule at all. It may be the answer to which buildings have anything to gain — a question

an individual building cannot answer about itself, since to know that 41 % is available one must have seen buildings where 2 % is available.

11.1 The fund as a way of finding those buildings

The second of these was tested directly. Fifteen predecessors in each of two populations contribute records; the median of those records is the fund. Twenty newcomers, none of them in the fund, each receive the fund and forty of their own evaluations. Their gain is compared with their own ceiling, measured separately.

	Across 20 newcomers
ceiling	+4.1 % (present in 100 %)
fund, unchanged	-0.0 % (helped 45 %)
fund, refined	+0.6 % (helped 85 %)
correlation of fund gain with ceiling	+0.91 (rank +0.81)

Table 12.

Table 12. What the fund takes of what is available, and how closely the two are related.

Two groups separate. In the nine newcomers whose ceiling exceeds 5 %, the ceiling has a median of 7.5 % and the fund with refinement returns 4.3 %. In the eleven below 5 %, the ceiling is 1.1 % and the fund returns 0.2 %. In the two largest cases the fund recovered 74 % and 53 % of what was available.

The correlation of +0.91 is the point. A building's gain from the fund tracks how much that building had to gain. So the fund can be read as a filter. Ranking the twenty newcomers by their gain and taking the first five captures 77 % of all the gain available in the population; five buildings chosen at random capture 30 %.

Twenty newcomers in two populations is a small sample and the correlation is measured within one synthetic domain. But the mechanism does not depend on the domain. A fund answers a question no individual can answer about itself, and the cheapest useful form of that answer is a ranking: which objects are worth visiting, and which are already close to their own limit.

The same measurement calibrates the platform against reality, and the result is not flattering. The median of 2.5 % is the whole range of the schedule game in this model. Measured re-commissioning of existing buildings gives a median of 16 % of whole-building energy across 643 buildings with a payback of 1.1 years [10]. Advanced controls retrofitted to 66 rooftop units gave 22–90 % of unit energy, averaging 57 % [11]. One campus programme reported 43 % in a single building from correcting its schedule alone [12]. Real waste sits in things this model does not contain: plant running in an empty building, a stuck economiser, a fan at full speed. The model has one zone and a setpoint.

The platform compares conditions of transfer honestly against each other. It understates by a large factor how much is available in a real building.

12. What the steps say together

- An inherited representation used unchanged is the worst of the conditions tested, in every domain.
- A single predecessor is unreliable and cannot be chosen in advance by any obvious measure of similarity, including physical parameters, floor area, year and declared use.
- The quality a predecessor achieved does not pass to its successors.
- What survives transfer is the arrangement of observation and regulation — where to watch, what to treat as a signal — rather than the values an individual arrived at.
- Inheritance from a single predecessor becomes actively harmful where the event is rare: in an ordinary year such a predecessor helps about half of newcomers, and when rare conditions are present it helps none, at both integration steps (§4). Where many events are needed before an object can judge at all, an inherited rule is the only source available (§6.2).
- Quantity of records converts into value only through relations between conditions and settings, and those relations must be paid for in records. The first hundred are worth more than the next hundred.
- A record must carry the state of the object, not only its nameplate, or accumulation amplifies harm.
- Representation dominates. The same records expressed relative to the receiver almost cease to do harm.
- What a fund gives an object is closely related to how much that object had to gain (correlation +0.91). A fund can therefore be used to rank objects by how much is available in them, which no object can determine about itself.

For the general question the results give one working answer: what is worth inheriting is what is expensive to acquire again, written in terms that refer to the receiver rather than to the donor.

13. Limitations

- Three domains, two of them simulated. The measured domain contains no setpoints and no record of operator intervention, so the inherited object there is a predictive model rather than a policy.
- The range of the synthetic domain is narrow: a median of 2.5 % against 16 % measured in real re-commissioning. Orderings between conditions should be read; magnitudes should not be carried over to real buildings.
- The one-hour integration step produces a temperature oscillation of several degrees in every run. Its effect was measured for §4, where the ordering of conditions is unchanged at ten minutes; the other experiments were not repeated at the finer step, so their magnitudes should not be relied on.
- Cost weights, the ranges of body parameters and the frequency of rare conditions are choices of the author. With different choices the magnitudes, though probably not the ordering, would change.
- The rare-condition result for a damaged object rests on eight cases and must be repeated.

- The heritability slope is negative in two populations but its magnitude differs fivefold; only the absence of a relation is established.
- The ranking result of §11.1 rests on twenty newcomers in one synthetic domain and has not been tested on measured data.
- The platform measures one function — inheritance — in a framework that names several others: retention, restoration, selection, mutation. Their protocols are defined and not yet run.
- Benchmark water networks are distribution systems, not the internal pipework of a single building. The mechanism carries over. The numbers do not.

14. Reproducibility

All results are produced by open code with fixed random seeds, and every table is written by the script that produces it. The measured data are public [4]. The water networks ship with WNTR [6]. Total computation is a few hours on an ordinary laptop.

The experiments were designed by the author. The code and the manuscript were prepared with AI assistance (Claude, Anthropic). The questions, the protocol, the choice of what to measure, the decision of what to report, and the interpretation of the results are the author's. No AI system is an author, and the author takes full responsibility for the content.

15. Where to begin

Three things would change the answers, in order of how much they would change them.

Data with setpoints and a log of interventions. The measured domain here has meter readings and nothing else, which is why the inherited object had to be a model rather than a policy. Sources exist: instrumented reference buildings, and thermostat data donated by householders, of which one programme covers more than 200 000 homes [13].

A ten-minute integration step, and everything re-run.

An object with a described structure of regulation — a set of observed quantities, rules of response, priorities — rather than a vector of numbers. Every experiment here transferred values. The results say values are the wrong thing to transfer. The next experiment should transfer the arrangement, and the first task is to describe an object that has one.

References

- [1] Xu, S., Wang, Y., Wang, Y., O'Neill, Z., Zhu, Q. One for Many: Transfer Learning for Building HVAC Control. Proc. 7th ACM Int. Conf. on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '20), 2020. doi:10.1145/3408308.3427617; arXiv:2008.03625.
- [2] Pinto, G., Wang, Z., Roy, A., Hong, T., Capozzoli, A. Transfer learning for smart buildings: a critical review of algorithms, applications, and future perspectives. Advances in Applied Energy 5, 100084 (2022). doi:10.1016/j.adapen.2022.100084.
- [3] A HOT Dataset: 150 000 Buildings for HVAC Operations Transfer Research. Proc. 12th ACM Int. Conf. on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '25), 2025. doi:10.1145/3736425.3770110.

- [4] Miller, C., Kathirgamanathan, A., Picchetti, B. et al. The Building Data Genome Project 2, energy meter data from the ASHRAE Great Energy Predictor III competition. *Scientific Data* 7, 368 (2020). doi:10.1038/s41597-020-00712-x.
- [5] Rossman, L. A. EPANET 2 Users Manual. U.S. Environmental Protection Agency, 2000.
- [6] Klise, K. A., Bynum, M., Moriarty, D., Murray, R. A software framework for assessing the resilience of drinking water systems to disasters with an example earthquake case study. *Environmental Modelling & Software* 95, 420–431 (2017).
- [7] A rule-based water quality sensor placement method for water supply systems using network topology. *Water Resources Management*, 2023. doi:10.1007/s11269-023-03685-9.
- [8] Strategic sensor placement using centrality metrics and machine learning techniques for leak localization in water distribution networks. *Water Supply* 26 (2), 98 (2026). doi:10.2166/ws.2026.006.
- [9] Evaluation of retrocommissioning persistence in large commercial buildings. National Conference on Building Commissioning, 2004. Lawrence Berkeley National Laboratory.
- [10] Mills, E. Building commissioning: a golden opportunity for reducing energy costs and greenhouse gas emissions in the United States. *Energy Efficiency* 4, 145–173 (2011). doi:10.1007/s12053-011-9116-8.
- [11] Wang, W., Katipamula, S., Ngo, H. et al. Advanced Rooftop Control (ARC) Retrofit: Field-Test Results. PNNL-22656, Pacific Northwest National Laboratory, 2013.
- [12] Small Workplace Automation and Remote Monitoring (SWARM). University of California, Davis, Facilities Management.
- [13] Donate Your Data. ecobee. Anonymised thermostat records from more than 200 000 homes, available to researchers on application.
- [14] Luo, J., Miras, K., Tomczak, J., Eiben, A. E. Enhancing robot evolution through Lamarckian principles. *Scientific Reports* 13, 21109 (2023). doi:10.1038/s41598-023-48338-4.